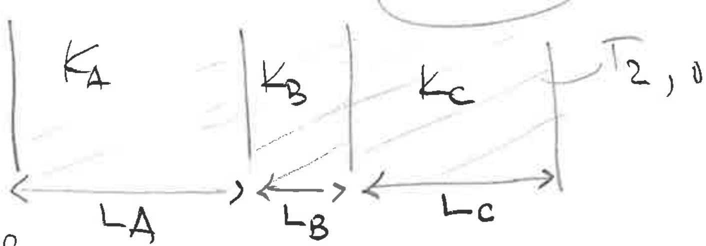
3.16, 3.43, 4.28, 4.35

Homework #4

Chem 312
Spring 2020
Folku Adobue

97/100

3.16



We have,

outer surface Temp $= T_{2,0} = 20°C$

Inner surface Temp $= T_{1,i} = 600°C$

Oven Air temp $= T_\infty = 800°C$ $\qquad k_B = ?$

$L_A = 0.40m \qquad L_C = 0.20m \qquad k_C = 60 W/m.K$

$L_B = 0.20m \qquad K_A = 25 W/m.K$

$h = 25 W/m^2.K$

$$q = h(T_\infty - T_{1,w}) = 25(800-600) = \underline{5000 W/m^2}$$

25/25

$$q = \frac{(\Delta t)\text{overall}}{(R_{th})\text{total}}$$

$$q = \frac{T_\infty - T_{2,0}}{(R_{th})_{conv,\infty} + R_{th-A} + R_{th-B} + R_{th-C}}$$

$$5000 = \frac{800-20}{\frac{1}{h_{air}} + \frac{L_A}{k_A} + \frac{L_B}{k_B} + \frac{L_C}{k_C}}$$

$$5000 = \frac{800-20}{\frac{1}{25} + \frac{0.4}{25} + \frac{0.2}{k_B} + \frac{0.20}{60}} = 5000 \times \left(\frac{1}{25} + \frac{0.4}{25} + \frac{0.2}{k_B} + \frac{0.2}{60}\right) = \underline{780}$$

$$\frac{1}{25} + \frac{0.4}{25} + \frac{0.2}{k_B} + \frac{0.2}{60} = \frac{780}{5000} \Rightarrow \frac{0.2}{k_B} = \frac{780}{5000} - \left(\frac{1}{25} + \frac{0.4}{25} + \frac{0.2}{60}\right)$$

$$\frac{0.2}{k_B} = 0.156 - 0.05933$$

$$k_B = \frac{0.2}{0.156 - 0.05933}$$

$$k_B = 2.0689 \; W/m.K$$

1 · $4 = 3.14\,(6\times10^{-3}) \times 1.25\,(T_{so} - T_\infty) + \pi\, d_o \sigma\,(T_{so}^4 - T_{sur}^4)\,\varepsilon_i$

$4 = 3.14\,(6\times10^{-3})\times1.25\left[\frac{T_{so}-293}{6\times10^{-3}}\right]^{\frac{1}{4}}(T_{so}-293) + \pi\,(6\times10^{-3})$

$\times\, 5.67\times10^{-8} \times (T_{so}^4 - 293^4) \times 0.9$

$T_{so} = 307.83\,K = 34.83\,°C$ as outer temperature of surface of insulation

Conduction through insulation

$$q' = \frac{T_{si} - T_{so}}{R_{th\,cond}} \Rightarrow R_{th\,cond} = \frac{\ln(r_2/r_1)}{2\pi\,(0.25)}$$

T_{ki}

$$R_{th\,cond} = \frac{\ln(3/1)}{2\pi\,(0.25)} = 0.6994$$

$$q' = 4\,\frac{W}{m} = \frac{T_{si} - 307.83}{0.6994}$$

$T_{si} = 2.7976 + 307.83$

$= 310.63\,K$

$T_{si} = 37.63\,°C$

Inner surface temp of insulation

Write a couple of lines inferring the effects of insulation

4.28

We know:

$D = 0.25\,m$ $\quad L = 2\,m$ $\quad T_{\infty,2} = 25\,°C$

$W = 1\,m$ $\quad k = 150\,W/mK$ $\quad T_{\infty,1} = 300\,°C$

$h_1 = 50\,W/m^2K$ $\quad h_2 = 4\,W/m^2K$

(25/25)

$$q = \frac{T_{\infty,1} - T_{\infty,2}}{R_{conv,1} + R_{cond} + R_{conv,2}}$$

Total rate of heat

4.35

* With the horizontal boundary and the vertical boundary

$\Downarrow$

Convection process

So, Energy balance

$E_{in} - E_{out} = 0$

$q_1 + q_2 + q_3 + q_4 + q_5 + q_6 = 0$

$$k(\Delta y \cdot 1)\frac{T_{m-1,n} - T_{m,n}}{} + k\left[\frac{\Delta x}{2} \cdot 1\right]\frac{T_{m,n-1} - T_{m,n}}{\Delta y} + h\left[\frac{\Delta y}{2} \cdot 1\right](T_\infty - T_{m,n})$$

$$+ 0 + k\left[\frac{\Delta y}{2} \cdot 1\right]\frac{T_{m+1,n} - T_{m,n}}{\Delta x} + k(\Delta x \cdot 1)\frac{T_{m,n+1} - T_{m,n}}{\Delta y} = 0$$

$\left(\frac{23}{25}\right)$

considering

$\Delta x = \Delta y$ & regrouping,

$$\boxed{2(T_{m-1,n} + T_{m,n+1}) + (T_{m+1,n} + T_{m,n-1}) + \frac{h\Delta x}{k}T_\infty - \left[6 + \frac{h\Delta x}{k}\right]T_{m,n} = 0}$$

should be 4

* Considering both boundary insulated:

$q_3 = q_4 = 0$ {why?}

we will

$$\boxed{2(T_{m-1,n} + T_{m,n+1}) + (T_{m+1,n} + T_{m,n-1}) - 6T_{m,n} = 0}$$

✗